Analysis of SDLC Models Applied to a Health Management System

Aim

To comprehensively analyze the Software Development Life Cycle (SDLC) models applicable to the development of a Health Management System (HMS) and identify the most optimal framework for its successful implementation.

Objective

- Study and evaluate diverse SDLC methodologies in the context of healthcare technology.
- Analyze the unique development requirements of a Health Management System, including compliance and data security
- .Compare the advantages and limitations of various SDLC models for this specific domain.
- Recommend the most strategic and efficient SDLC model for the HMS application.

Problem Statement

Healthcare systems mandate rigorous regulatory compliance (e.g., HIPAA), uncompromising data security, high availability, and the ability to adapt to evolving medical standards and patient needs. Traditional, rigid software development methodologies often fail to accommodate these dynamic shifts, resulting in delayed deployments, budget overruns, and systems that struggle to integrate with modern clinical workflows.

Solution

Implementing an Agile-driven SDLC, augmented by DevOps practices, serves as the optimal solution. This approach facilitates iterative development, allowing continuous integration of stakeholder (clinician and patient) feedback. Furthermore, it ensures rapid deployment of critical security patches, seamless feature enhancements, and steadfast adaptability to healthcare regulatory changes without disrupting existing operational stability.

System Overview: Health Management System (HMS)

A comprehensive Health Management System empowers stakeholders by centralizing healthcare delivery operations.

Key functionalities include:

- Secure patient registration and medical record management.
- Real-time appointment scheduling and tele-consultation facilitation.
- Electronic Health Record (EHR) and E-Prescription management for medical professionals.
- Automated billing, insurance claim processing, and institutional compliance reporting.

Core Functional Requirements

- Patient Module
  User Registration & Authentication (Multi-Factor Authentication)
  Search & Book Consultations
  Access to Personal Medical History & Lab Reports
  Secure Online Payments
- Healthcare Provider Module
  Schedule & Appointment Management
  Access and Update Patient EHRs
  Generate E-Prescriptions
  Telemedicine Interface Integration
- Administrative Module
  User & Role Management (Access Control)
  Billing & Insurance Processing
  Inventory & Resource Management
  Analytics & Compliance Reporting

Conclusion

This case study underscores that the selection of an SDLC model is intrinsically tied to the operational criticality and domain-specific requirements of the project. For a dynamic, high-stakes application like a Health Management System, traditional linear models fall short. Integrating an Agile framework with DevOps practices affords the essential flexibility, continuous stakeholder collaboration, and rapid, secure deployment capabilities required. This synergistic approach guarantees the delivery of a resilient, compliant, and highly functional health management platform that consistently meets the evolving demands of the healthcare ecosystem.